Kelechi Emeka Ogbonna

Biotechnologist who builds computer simulations

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ABOUT ME

I trained as a biotechnologist and now write software that simulates how things behave. Examples include diseases in the body, an AI landing a rocket, and space programmes choosing what to build. I care about testing my work properly and saying clearly what a result does and does not prove. I am looking for an internship or graduate role in computer modelling, space life sciences or simulation.

PROJECTS

Teaching an AI to land a rocket — 2026

- Trained an AI to land a rocket in a physics simulator. It landed safely on the pad **499 times out of 500**; a hand-written control program managed 399.
- Showed how a badly designed reward makes an AI "cheat": paid for staying upright, it learned to hover forever instead of landing. I fixed the reward, and landings returned to 98%.
- Tested the AI in strong wind and heavier gravity it had never seen, and reported where it failed as well as where it succeeded.

When should a space programme abandon a plan? — 2026

- Built a simulation of a space programme choosing which technologies to develop for the Moon, Mars and beyond. Some paths secretly lead nowhere.
- Compared six decision strategies across 400 simulated worlds. The best was "commit fully, but quit a project once the evidence shows it is failing". It achieved **16% more progress** and **cut wasted budget by more than half** compared with never quitting.
- Found and corrected a mistake in my own model, then reran every experiment before reporting results.

Modelling how cancer cells use energy (Project Confluence)

- Wrote a mathematical model of cancer-cell metabolism and tested it against real laboratory data from 928 cancer cell lines.
- Using real data instead of made-up data let the model pin down **15 of its 17 unknowns**, up from 7. The software is about 43,000 lines of Python, checked by more than 330 automated tests.

Other projects

- **Breast cancer energy model** (M.Sc. research): simulated how aggressive breast-cancer cells handle energy and stress.
- **Fly Brain vs. Tumor**: a browser game where a simplified fruit-fly brain learns to dose a simulated tumour.
- **simplifyr**: a payments app built at a 2026 hackathon (Solana Mobile track).

RESEARCH WRITING

Around 50 independent research papers (not peer-reviewed) on modelling disease and astronaut health, collected at research-theses-hub. Highlights:

- **AI-assisted planning of cervical cancer care**, published on Zenodo (doi:10.5281/zenodo.23210715).
- **Astronaut health**: bone loss on a trip to Mars, radiation damage from solar storms, and how medicines degrade on the Moon.

EDUCATION

M.Sc. Biotechnology · Ahmadu Bello University, Zaria — In progress

B.Sc. Biotechnology (Hons) · Nile University of Nigeria, Abuja — 2022
Final project: made silver nanoparticles from papaya leaves and tested them in the lab as a possible diabetes treatment.

EXPERIENCE

Biotechnology Instructor · NYSC, Etinan College of Health and Technology

- Taught biotechnology courses and ran laboratory classes; introduced students to simple mathematical models of biology.

Research Intern · National Biotechnology Development Agency, Abuja

- Worked on DNA sequencing and lab analysis for genetically modified crop projects.

Laboratory Intern · Arubah Medical Diagnostic Services, Calabar

- Tested patient samples using PCR and other standard lab methods.

Managing Director · Uloma Shopping Mall | **POS Operator** · MoneyPoint Microfinance Bank

- Ran a retail business day to day (staff, stock, suppliers and money) and handled customer payments at a microfinance bank.

SKILLS

Programming	Python (including machine learning and scientific libraries), TypeScript, MATLAB, Git
Methods	Computer simulation, mathematical modelling, machine learning, statistics
Laboratory	PCR, DNA sequencing, spectrophotometry, nanoparticle synthesis
Other	IELTS 7.5 · Google Project Management certificate · Anthropic AI courses